Marmara Üniversitesi
Electrical and Electronics Engineering Department
PHYS 1101: Physics I (Fall 2021)

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Course Information	**Textbook**: Fundamentals of Physics, Halliday/Resnick, 10th Ed.		Grading	Class Work 20%	Mid-term Exam 40%	Final Exam 40%

Week	Events	Topics to cover	Ch	Sec	Exercises
1 Oct 4-8	---	**Measurements and Units**. SI units and conversions, Dimensional analysis, Uncertainty Analysis.	1	1, 2, 3	(**1**) 1,9,12,13,14,21,22,27,30.
2 Oct 1-5	---	**Motion in a Straight Line**. Definitions, Instantaneous velocity, equation of motion for constant acceleration, Free-fall Analysis.	2	1, 2, 3, 4, 5	(**2**) 1,4,5,7,11,15,17,19,22, 23,25,27,35,37,41,44,50,52.
3 Oct 18-22	Drop/Add week	**Vector Analysis**. Vector operations, Unit vectors and decompositions, Dot and cross products.	3	1, 2, 3	(**3**) 2,3,9,17,24,33,36,43.
4 Oct 25-29	Cumhuriyet Bayramı Oct 29	**Motion in 2D and 3D**. Position, Displacement, Average and instantaneous velocity, Average and instantaneous acceleration, projectile motion, uniform circular motion.	4	1, 2, 3, 4, 5	(**4**) 3,6,7,11,14,16,22,27,28, 56,58,66,73,80.
5 Nov 1-5	Quiz 1 Nov 3	**Force and Motion I**. Newton's laws of motion, Gravitation, free-fall and weight, Applications.	5	1, 2, 3	(**5**) 2,4,6,10,12,13,17,21,27, 34,39,56,57.
6 Nov 8-12	---	**Force and Motion II**. Friction, Drag and terminal velocity, circular motion.	6	1, 2, 3	(**6**) 1,5,7,10,13,17,20,25,29, 36,39,42,44,47,57.
7 Nov 15-19	---	**Kinetic Energy and Work**. Kinetic energy, Work, Gravitational work, Work by a spring, Power.	7	1, 2, 3, 4, 5, 6	(**7**) 1,5,7,11,14,15,16,19,23, 27,29,34,35,36,41,43,46,50.
Nov 20-28	**Mid-term Exams period**				
8 Nov 29-Dec 3	---	**Conservation of Energy**. Potential energy, Conservation of mechanical energy, Work by external forces, Conservation of energy.	8	1, 2, 3, 4, 5	(**8**) 1,2,5,7,9,13,15,16,23,29, 42,48,49,53,57.
9 Dec 6-10	---	**Center of Mass and Linear Momentum**. Center of mass, System of particles, Linear momentum.	9	1, 2, 3	(**9**) 1,2,3,4,9,10,14,18,20,25.
10 Dec 13-17	---	**Center of Mass and Linear Momentum**. Collision and impulse, Conservation of linear momentum, momentum and kinetic energy, Elastic collisions.	9	4, 5, 6, 7	(**9**) 27,32,35,38,46,49,50,52.
11 Dec 20-24	---	**Rotation**. Definitions, Rotation with constant angular acceleration, linear and angular variables, kinetic energy of rotation, Inertia, Torque, Newton's laws for rotation, Work.	10	1-8	(**10**) 2,4,7,9,10,11,14,19, 22, 23,34,36,46,48,50,51,58,60.
12 Dec 27-31	New Year 2022	**Angular Momentum**. Torque, Angular momentum, Newton's second law in angular form.	11	4, 5, 6	(**11**) 19,21,25,26,30,32, 35,36,37.
13 Jan 3-7	Quiz 2 Due Jan 5	**Angular Momentum**. Angular momentum of a rigid body, Conservation of angular momentum.	11	7, 8	(**11**) 40,49,50,59,60.
14 Jan 10-14	---	**Static Equilibrium and Elasticity**. Static equilibrium, Examples, Elasticity.	12	1, 2, 3	(**12**) 2,5,10,15,21,23,31,33, 38,43,44,48.
15 Jan 17-21	---	**Review**	---	---	---
Jan 24- Feb 6	**Final Exams period**	**Feb 8-14 (Re-sit Final Exams period)**			